

Cholesterol Metabolism Gene Signatures Distinguish Estrogen-Receptor Status and Define Immune-Relevant Molecular Subtypes of Breast Cancer

An Integrated Multi-Cohort Analysis of TCGA-BRCA and METABRIC

Abstract

Purpose. Cholesterol metabolism is closely linked to estrogen-receptor (ER) biology, but its value for breast cancer classification, molecular subtyping and causal inference remains incompletely quantified.

Materials and Methods. We applied co-expression network screening, machine-learning feature selection, consensus clustering, single-cell localization, immune deconvolution (CIBERSORT/ESTIMATE), SMR/HEIDI Mendelian randomization and Bayesian colocalisation to TCGA-BRCA (n = 952) and METABRIC (n = 1,979), with classifier validation in GSE21653 and GSE7390 and nearest-centroid subtype mapping across all external cohorts.

Results. The strict WGCNA-by-pathway intersection was empty. A pathway-first adaptation using 119 measurable cholesterol genes yielded 11 hub genes; the ER classifier reached AUCs of 0.946 (5-fold cross-validation), 0.922 (METABRIC), 0.847 (GSE21653) and 0.903 (GSE7390). Consensus clustering defined four subtypes with divergent ER composition, immune infiltration and PAM50 correspondence. Single-cell analyses localized cholesterol biosynthesis to tumour epithelial and myeloid/endothelial compartments. Pathway-wide SMR identified FDPS as a putatively causal protective gene ($P = 9.8e-7$; HEIDI $P = 0.026$; coloc.abf PP.H4 = 0.9985), with the protective direction replicated in ER+ OncoArray analyses ($P = 1.54e-5$) and an inverse methylation-expression correlation (ER-adjusted partial $\rho = -0.187$, $P = 2.2e-6$).

Conclusion. Cholesterol metabolism genes do not define a progression-associated co-expression module but carry a strong, cross-platform ER signature, define immune-relevant molecular subtypes and include FDPS as a putatively causal protective gene for breast cancer risk.

Keywords: Breast Neoplasms; Cholesterol; Machine Learning; Receptors, Estrogen; Tumor Microenvironment

1. Introduction

Cholesterol and its derivatives support breast cancer cell proliferation, membrane biogenesis, steroid hormone synthesis and immune regulation [1,2]. ER-positive (ER+) tumours coordinately up-regulate cholesterol biosynthetic enzymes, and cholesterol-lowering agents have been proposed as adjunct

therapy [3-6]. Most lipid-metabolism transcriptomic studies have focused on prognosis [7,8]; here we additionally report an honest null network-pathway intersection, an ER-focused classifier validated on three platforms, and a germline causal-inference layer.

We applied a five-layer bioinformatics framework - co-expression network screening, machine-learning feature selection, patient-level molecular subtyping, single-cell localization and Mendelian-randomization causal testing - to cholesterol metabolism in breast cancer, using TCGA-BRCA and METABRIC as discovery and external cohorts.

2. Methods

2.1 Data sources and verification The overall study design and the flow of analyses are summarised in Supplementary Figure S2. For validation analyses, PAM50 calls for TCGA-BRCA were obtained from UCSC Xena (GDC hub) [9]; FDPS HM450 methylation from the cBioPortal data repository (brca_tcga_methylation_hm450); the independent expression cohorts GSE21653 (Affymetrix HG-U133 Plus 2.0, NCBI GEO with GPL570 annotation) [10] and GSE7390 (Affymetrix HG-U133A, ArrayExpress E-GEOD-7390 processed matrix and SDRF clinical data, probes mapped with hgu133a.db) [11] were used; and the second single-cell atlas GSE161529 (Mendeley Data mirror of the Pal et al. atlas) [12] was used for single-cell validation. TCGA-BRCA RNA-sequencing STAR counts, methylation, mutation and clinical files were obtained from the NCI GDC portal [13] and verified against the official manifest by MD5 checksums (4,393/4,393 files); progression-free interval (PFI) was taken from TCGA-CDR [14]. METABRIC expression and clinical annotation with recurrence-free survival (RFS) were obtained from cBioPortal [15].

2.2 Cohorts and endpoints

The discovery cohort comprised 952 TCGA-BRCA patients [16] with valid PFI (122 events, 12.8%; median follow-up 2.1 years; ER status for 909). The external cohort comprised 1,979 METABRIC samples [17] with RFS (803 events, 40.6%; median follow-up 8.5 years; ER status for all).

2.3 Cholesterol metabolism gene set

A cholesterol metabolism gene set was assembled from KEGG hsa00100 (steroid biosynthesis) [18] and Gene Ontology biological-process terms GO:0008203 (cholesterol metabolic process) and GO:0006695 (cholesterol biosynthetic process) [19] using org.Hs.eg.db [20]; the union contained 120 unique genes (19 from KEGG, 107 from GO:0008203, 39 from GO:0006695).

2.4 WGCNA

Expression was transformed as $\log_2(\text{CPM}+1)$. The 8,000 most variable genes entered weighted correlation network analysis [21,22]: soft-threshold power selection by scale-free fit, topological overlap matrix, average-linkage hierarchical clustering and merging of modules with eigengene correlation ≥ 0.80 . Module eigengenes were correlated with PFI, ER status and stage with Benjamini-Hochberg FDR [23].

2.5 Candidate selection and machine learning The strict intersection of trait-associated modules with the cholesterol set was empty; the candidate pool was therefore the 119 cholesterol genes measurable in both cohorts (pathway-first adaptation, documented as an analysis deviation). LASSO logistic regression [24] and SVM-RFE (linear SVM) [25] were applied to ER status; the intersection of their selected lists defined 11 hub genes. Selection stability was assessed with 300 bootstrap resamples of the LASSO fit. A signature score was computed as the sum of LASSO coefficients times z-scored expression. Performance was assessed by 5-fold cross-validation (CV) AUC in TCGA and external AUC in METABRIC (2,000-bootstrap 95% confidence intervals); calibration was evaluated with reliability curves, Brier scores and logistic recalibration; clinical utility with decision-curve analysis.

The strict intersection of module genes with the cholesterol set was empty; the candidate pool was therefore defined as the 119 cholesterol genes measurable in both cohorts (pathway-first adaptation, documented as an analysis deviation). Expression was z-scored with TCGA statistics. LASSO logistic regression [24] (L1, C selected by 5-fold CV over a 25-point grid) and SVM-RFE (linear SVM) [25] were applied to ER status; the intersection of their selected gene lists defined 11 hub genes. A signature score was computed as the sum of LASSO coefficients times z-scored expression. Performance was assessed by 5-fold CV AUC in TCGA and external AUC in METABRIC with 2,000-bootstrap 95% confidence intervals. As a secondary endpoint, the association of the signature with PFI/RFS was tested by univariate Cox regression.

2.6 Molecular subtyping and characterization

Consensus clustering [26] (100 resamples, 80% of patients, k-means, $k = 2-6$) was applied to hub-gene z-scores; the number of clusters was selected by the proportion of ambiguous clustering (PAC). Subtypes were compared for ER composition, survival, immune marker scores (nine curated cell-type panels), the cholesterol-pathway score, differential expression (Mann-Whitney, BH-FDR) and KEGG/GO enrichment of strong-effect genes. TCGA subtype centroids were mapped onto METABRIC, GSE21653 and GSE7390 by nearest-centroid assignment. Immune programs were quantified by single-sample GSEA (ssGSEA) [27], CIBERSORT (nu-SVR, LM22, 22 immune cell types) [28] and ESTIMATE (immune/stromal scores) [29]; immune-checkpoint genes (CD274, PDCD1, CTLA4, LAG3, HAVCR2, IDO1, CD8A) were compared across subtypes. Hub-gene and FDPS expression were compared across PAM50 intrinsic subtypes.

2.7 Software

Python 3.11 (scikit-learn [30], scipy, lifelines [31], pandas, numpy, matplotlib) and R 4.6.1 (org.Hs.eg.db, hgu133a.db, susieR, coloc). All random seeds fixed at 20260804.

2.8 Single-cell localization and SMR causal inference

Single-cell localization used the processed GSE176078 atlas [32] (100,064 cells; CELLxGENE release; three annotation resolutions); hub-gene mean expression, detection percentage and Wilcoxon enrichment (BH-FDR) were computed per cell type and confirmed in GSE161529 [12]. Causal inference used SMR/HEIDI v1.3.1 [33,34] with eQTLGen whole-blood cis-eQTLs [35] (31,684 individuals), the BCAC overall and triple-negative breast cancer GWAS [36] (247,173 and 118,987 women) and the 1000 Genomes Phase 3 European LD reference, scanning 119 cholesterol genes (251 probes) with a +/-2 Mb cis window, instrument eQTL $P < 5e-8$ and Bonferroni $\alpha = 0.05/119$. ER-stratified sensitivity used the BCAC OncoArray public release [37] (ER+ 69,501; ER- 21,468 cases). GTEx v8 breast-mammary and whole-blood SMR-format BESDs [38,39] were used for tissue-specific and replication analyses. For FDPS, single-causal-variant Bayesian colocalisation (coloc.abf; $p_1 = p_2 = 1e-4$, $p_{12} = 1e-5$, $W = 0.2$) and multi-signal SuSiE/coloc.susie [40,41] were applied; detailed parameter settings are provided in the archived code.

3. Results

3.1 Cohorts

TCGA-BRCA contributed 952 patients (122 PFI events; median follow-up 2.1 years; ER+ 699/909). METABRIC contributed 1,979 samples (803 RFS events; median follow-up 8.5 years; ER+ 1,505/1,979). Both cohorts covered 19,448 shared genes.

Table 1. Cohorts used in this study.

Characteristic	TCGA-BRCA	METABRIC
Patients, n	952	1,979
Events, n (%)	122 (12.8)	803 (40.6)
Median follow-up, years	2.1	8.5
ER+ / ER-, n	699 / 210 (43 unknown)	1505 / 474
Stage I / II / III / IV, n	170 / 543 / 209 / 19	500 / 825 / 118 / 10

3.2 WGCNA and the null module-by-pathway intersection

At soft-threshold power $\beta = 5$ (scale-free fit $R^2 = 0.834$), the 8,000-gene network resolved into 15 modules (38-853 genes). The module most strongly associated with PFI events (M2238; 69 genes) showed a weak correlation ($r = 0.082$, $P = 0.012$) that did not survive multiple-testing correction (BH-FDR

= 0.170), whereas ER status showed very strong module associations (M2314; $r = -0.794$, BH-FDR $\sim 4e-197$).

The strict intersection of trait-associated modules with the 120-gene cholesterol set was empty: the PFI module (M2238) and the ER module (M2314) contained no cholesterol genes. No cholesterol gene was significantly associated with PFI (smallest Cox $P = 0.13$, CH25H), whereas several strongly tracked ER status (LIPE $P = 4.5e-17$; EBP $P = 2.4e-12$; FDXR $P = 6.0e-12$). This honest null result motivated the pathway-first adaptation below.

Table 2. Selected module-trait associations (WGCNA, 15 modules; BH-FDR per trait).

Module	Size	Trait	r	P	BH-FDR
M2238	69	PFI event	+0.082	0.0117	0.170
M3105	853	PFI event	-0.071	0.0274	0.170
M2722	330	PFI event	+0.069	0.0340	0.170
M2314	157	ER status	-0.794	2.7e-198	4.1e-197
M2238	69	ER status	-0.612	1.1e-94	7.9e-94
M1703	105	ER status	+0.514	1.9e-62	9.5e-62
M3448	254	PFI time	-0.114	4.1e-04	6.2e-03
M666	38	Stage	+0.104	1.4e-03	0.018

3.3 Hub genes and the ER-status classifier From 119 measurable cholesterol genes, LASSO ($C = 0.316$; 69 genes) and SVM-RFE (12 genes) selected 11 hub genes by intersection (Table 3). In 300 LASSO bootstrap resamples all 11 hub genes were selected in at least 73.7% of fits (mean 90.2%; G6PD and VLDLR 100%, PRKAA1 99.7%, LIMA1 73.7%), supporting selection stability. The ER-status classifier reached internal 5-fold CV AUC 0.946 $\pm$ 0.012 (full-data 0.953) and external AUC 0.922 (95% CI 0.907-0.935) in METABRIC. Calibration was excellent in TCGA out-of-fold predictions (Brier 0.069; recalibration intercept 0.002, slope 1.06) and mildly underconfident in METABRIC (Brier 0.098; intercept 0.57, slope 1.41); decision-curve analysis showed positive net benefit across thresholds (Figure 3). The signature was not associated with PFI in TCGA (adjusted HR 0.90, $P = 0.151$) but showed an ER/stage-adjusted protective association with RFS in METABRIC (HR 0.90, 95% CI 0.82-0.98, $P = 0.013$).

Table 3. Hub genes (LASSO x SVM-RFE intersection) and their coefficients (ER+ vs ER-).

Gene	LASSO coefficient
VLDLR	-0.7441
G6PD	-0.7116
PRKAA1	-0.6345
DHCR7	-0.4016
NSDHL	-0.3827
HMGCS2	0.4234
HSD17B7	0.5089
FDXR	0.5122
LIMA1	0.6229
DHCR24	0.6322
ABCG1	0.6960

3.4 Molecular subtypes defined by hub genes Cross-cohort subtype statistics (sample size, ER composition, events and survival) are provided in Supplementary Table S3.

Consensus clustering selected $k = 4$ (PAC = 0.097). Subtypes differed markedly in ER composition (C1 13.6%, C2 93.8%, C3 97.5%, C4 60.1% ER+), signature score and cholesterol-pathway score (Table 4); TCGA survival differences were not significant (multivariate log-rank $P = 0.331$). Nearest-centroid mapping onto METABRIC reproduced the four subtypes with near-identical ER composition and significant survival differences (global log-rank $P = 7.8e-05$; C4 worst, C2 best), which attenuated after adjustment for ER status and stage (C4 vs C1 HR = 1.26, 95% CI 0.95-1.67, $P = 0.12$), indicating that the unadjusted differences largely tracked ER status and stage (Figure 7). Nearest-centroid mapping onto GSE21653 ($n = 252$) and GSE7390 ($n = 198$) reproduced the subtype ER gradient (Kruskal-Wallis $P = 4.2e-16$ and $P = 4.7e-16$, respectively); survival differences were significant in GSE21653 (global log-rank $P = 0.026$; C1 worst, C2 best) but not in GSE7390 ($P = 0.48$), consistent with ER-driven survival differences (Supplementary Figure S3).

Table 4. Subtype characteristics (TCGA-BRCA, $n = 952$).

Subtype	n	ER+ (%)	PFI events	Signature mean	Cholesterol score
C1	142	13.6	23	-4.39	-0.11
C2	352	93.8	41	1.11	0.02
C3	296	97.5	34	1.73	-0.03
C4	162	60.1	24	-1.72	0.12

3.5 Immune microenvironment and pathway programs across subtypes

All nine immune/stromal marker scores differed across subtypes (Kruskal-Wallis $P < 0.001$): C1 showed the highest CD8 T, regulatory T, NK, B-cell and macrophage scores and the lowest stromal score; C3 the lowest across most immune populations; C2 the highest dendritic-cell and stromal scores; C4 the highest neutrophil score; the pattern reproduced in METABRIC (Supplementary Figure S1). Strong-effect differential genes ($|\text{mean log}_2 \text{ difference}| \geq 1$: 1,840/304/225/92 for C1-C4) showed C1-enriched cell-cycle programs and C3-enriched oxidative-phosphorylation and ribosomal programs. ssGSEA confirmed these patterns (all Kruskal-Wallis $P < 0.05$ except B cells, $P = 0.42$) with marker-score correlations of 0.40-0.93 (Figure 8; Supplementary Table S8). ESTIMATE confirmed the immune-hot/immune-cold axis (ImmuneScore highest in C1, Kruskal-Wallis $P = 1.9e-3$; StromalScore highest in C2, $P = 4.2e-22$; ESTIMATEScore $P = 3.5e-6$), and CIBERSORT showed significantly higher C1 fractions of M1 and M0 macrophages, activated dendritic cells, follicular helper T cells, monocytes and activated NK cells, with higher C2/C3 fractions of resting mast cells and M2 macrophages (all Kruskal-Wallis $P < 0.001$). Immune-checkpoint genes CD274, PDCD1, CTLA4, LAG3 and IDO1 were highest in C1 (Kruskal-Wallis $P = 1.7e-20$ to $4.9e-5$), while HAVCR2 and CD8A did not differ across subtypes ($P = 0.12$ and $P = 0.11$, respectively; Supplementary Tables S8 and S14).

Table 5. Immune/stromal marker scores by subtype (mean z; Kruskal-Wallis P).

Cell population	C1	C2	C3	C4	Kruskal-Wallis P
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CD8_T	0.316	-0.048	-0.116	0.041	7.78e-05
CD4_T	0.170	0.066	-0.194	0.062	1.37e-05
Treg	0.327	-0.011	-0.220	0.141	4.31e-11
NK	0.284	-0.039	-0.129	0.072	3.87e-05
B_cell	0.293	0.006	-0.131	-0.031	1.62e-04
Macrophage	0.118	0.039	-0.150	0.086	2.31e-04
DC	-0.042	0.179	-0.133	-0.108	8.37e-10
Neutrophil	0.145	-0.070	-0.111	0.228	1.57e-10
Stroma	-0.579	0.320	-0.134	0.056	2.18e-23

3.6 Single-cell localization of hub genes

In the 100,064-cell atlas, cholesterol biosynthetic enzymes (DHCR24, DHCR7, G6PD, HSD17B7, FDXR) were most strongly enriched in malignant epithelial cells (all Wilcoxon BH-FDR < 1e-10), while ABCG1, G6PD and LIMA1 were enriched in myeloid/endothelial cells and cancer-associated fibroblasts, localizing the pathway to both tumour and microenvironment compartments.

3.7 SMR causal inference

Pathway-wide SMR identified three probes passing the Bonferroni threshold in overall breast cancer: CYP51A1 ($P = 1.4e-8$), FDPS ($P = 9.8e-7$) and LSS ($P = 4.1e-5$). HEIDI rejected CYP51A1 and LSS as LD-confounded but supported FDPS ($P_{\text{HEIDI}} = 0.026$), with higher genetically predicted FDPS expression associated with lower breast cancer risk ($b = -0.42$, 95% CI -0.59 to -0.25); SREBF1, LIPA and FAXDC2 showed supportive protective signals. In the OncoArray ER-stratified sensitivity analysis the protective direction replicated (ER+ $P = 1.54e-05$, $b = -0.44$; ER- $P = 0.28$, $b = -0.15$, underpowered; lead SNP rs6677385), and the OncoArray overall analysis gave $P = 9.02e-06$. TNBC sensitivity found no Bonferroni-significant probe; G6PD, HMGCS2 and NSDHL lacked eQTLGen instruments. Per-probe results are provided in Supplementary Tables S2 and S10.

3.8 GTEx breast-tissue SMR sensitivity analysis

With GTEx v8 breast-mammary cis-eQTLs, 109/119 cholesterol genes had probes and 16 had genome-wide-significant instruments; no probe passed Bonferroni in overall or TNBC analyses. LIPA ($P = 8.2e-3$; $P_{\text{HEIDI}} = 0.53$) and FAXDC2 ($P = 1.15e-2$; $P_{\text{HEIDI}} = 0.69$) replicated direction-consistently with HEIDI support, matching the eQTLGen results. FDPS lacked a genome-wide-significant breast-tissue instrument, indicating that the blood-based FDPS signal is likely mediated through non-mammary tissues.

3.9 Colocalisation supports a shared causal variant at the FDPS locus

coloc.abf over 3,248 shared SNPs gave $PP.H4 = 0.9985$ at the FDPS locus (window sensitivity 0.996-0.997 at +/-1 Mb, +/-500 kb and +/-250 kb). The lead SNP rs12091730 was strongly associated with both FDPS expression ($P = 2.5e-18$) and breast cancer risk ($P = 1.8e-12$) with opposing directions, forming one LD block (11 SNPs, $r^2 \geq 0.8$). Single-SNP conditional analysis confirmed a single eQTL signal and additional GWAS-only signals (rs4971059, rs11264454) with no eQTL effects; sensitivity at $p_{12} = 1e-8$ gave $PP.H4 = 0.40$. However, multi-signal SuSiE/coloc.susie identified four eQTL credible sets and one GWAS credible set with maximum $PP.H4 = 0.44$ ($PP.H3 = 0.56$), so the single-shared-variant interpretation of coloc.abf should be regarded as suggestive (Figure 9; Supplementary Table S1). Bayesian colocalisation was extended to the three additional eQTLGen-supported genes (LIPA, FAXDC2, SREBF1); none showed

strong evidence of a shared causal variant with breast cancer risk (PP.H4 = 0.035, 0.043 and 0.171, respectively; PP.H1 dominated), indicating that their SMR signals are driven by strong eQTL associations without a corresponding GWAS signal (Supplementary Table S15).

3.10 FDPS expression and clinical features

FDPS expression was higher in ER-negative tumours (TCGA-BRCA $P = 5.7e-17$, Cohen's $d = -0.69$; METABRIC $P = 2.2e-48$, $d = -0.79$), unrelated to stage, and not independently prognostic after ER/stage adjustment (METABRIC adjusted HR = 1.03, $P = 0.48$). The germline eQTL-based causal signal is therefore distinct from the tumour-level association. In the Human Protein Atlas [42], FDPS showed cytoplasmic protein localization, tissue-enhanced expression in normal breast, and a non-significant TCGA protein-level prognosis ($P = 0.195$; HPA validation cohort $P = 0.023$, classified as unprognostic), consistent with the transcript-level null. FDPS mRNA was detected in all 17 cancer types with available cBioPortal RNA-sequencing data, indicating pan-cancer expression (Supplementary Table S13).

3.11 GTEx v8 whole-blood sensitivity analysis

No genome-wide-significant FDPS cis-eQTL was available in GTEx v8 whole blood ($n = 670$); under sample-size scaling from the eQTLGen instrument ($z = -9.65$, $N = 31,684$), the expected GTEx whole-blood z-statistic is -1.40 (29% power at $P < 0.05$; <0.01% at $P < 5e-8$), so this resource cannot independently instrument FDPS.

3.12 PAM50 calls recapitulate the hub-gene molecular subtypes

PAM50 calls were available for 871 TCGA-BRCA tumours. The hub-gene subtypes differed markedly in PAM50 composition (chi-square = 820.8, $P = 5.62e-168$; C1 86.4% Basal, C2 67.4% LumA, C3 48.6% LumA). Hub-gene and FDPS expression differed across all five PAM50 subtypes (Kruskal-Wallis $P < 0.001$ for all 12 genes), with FDPS highest in Basal and HER2-enriched tumours, consistent with its ER-negative enrichment (Figure 10; Supplementary Tables S4 and S9).

3.13 Independent validation in GSE21653

The ER classifier transferred to GSE21653 ($n = 252$, 83 disease-free-survival events) with AUC = 0.847 (95% CI 0.798-0.895; Brier 0.210), and FDPS remained strongly ER-associated (Mann-Whitney $P = 1.24e-10$; $d = -0.86$); FDPS survival associations attenuated after adjustment (HR = 1.16, $P = 0.250$). In a third independent cohort (GSE7390, $n = 198$, 91 RFS events), the classifier transferred with AUC = 0.903 (95% CI 0.848-0.949; Brier 0.119) and FDPS remained strongly ER-associated ($P = 4.78e-05$; $d = -0.68$), confirming cross-platform reproducibility (Figures 12 and 13; Supplementary Table S5).

3.14 Second single-cell atlas confirms FDPS enrichment in monocyte/macrophage populations

In GSE161529 (70,419 cells after QC), FDPS was most strongly expressed in monocyte/macrophage populations (Monocyte mean 0.65). T-myeloid ligand-receptor co-expression was detected for 7 of 20

curated pairs, strongest IL1B-IL1R2 (11.2% of T/myeloid cells), CXCL9/10/11-CXCR3 (2.0-2.3%) and CSF1-CSF1R (1.7%) (Figure 14; Supplementary Table S6).

3.15 FDPS DNA methylation is inversely associated with FDPS expression in TCGA-BRCA

In 672 TCGA-BRCA tumours with HM450 methylation and RNA-seq data, FDPS methylation correlated inversely with FDPS expression (Spearman $\rho = -0.203$, 95% CI -0.274 to -0.133, $P = 1.04e-07$). The association persisted after ER adjustment (partial $\rho = -0.187$, $P = 2.2e-6$) and in ER+ tumours ($\rho = -0.206$, $P = 4.6e-6$), with a consistent but non-significant direction in ER- tumours ($\rho = -0.142$, $P = 0.087$; $n = 147$) (Figures 15 and 16; Supplementary Table S7).

4. Discussion

4.1 Principal findings The five-layer framework was applied with real data. The strict WGCNA-by-pathway intersection was empty; the pathway-first adaptation produced an 11-gene signature that discriminated ER status with strong cross-platform performance (TCGA CV AUC 0.946; METABRIC 0.922; GSE21653 0.847; GSE7390 0.903). Consensus clustering defined four subtypes with divergent ER composition and immune infiltration, reproduced in METABRIC, GSE21653 and GSE7390. Single-cell analysis localized the signature to tumour epithelial and myeloid/endothelial compartments, and pathway-wide SMR identified FDPS as a putatively causal protective gene for breast cancer risk, supported by ER-stratified replication, HEIDI and coloc.abf (PP.H4 = 0.9985) but weaker under multi-signal SuSiE (max PP.H4 = 0.44). Tumour FDPS expression was strongly ER-associated and not independently prognostic.

GTEx v8 breast-mammary sensitivity directionally replicated LIPA and FAXDC2 but could not instrument FDPS in breast tissue; GTEx v8 whole blood lacked a significant FDPS cis-eQTL, consistent with the power available at $n = 670$.

4.2 Cholesterol metabolism and ER biology

The ER-status signal is biologically coherent: ER+ tumours up-regulate cholesterol biosynthetic enzymes to support steroid production and membrane turnover [3-5], and six hub genes are direct enzymes or regulators of cholesterol/steroid synthesis. The external AUC of 0.922 across platforms confirms that the signal is not a TCGA artifact. The near-null PFI associations indicate that cholesterol gene expression adds little independent prognostic information. FDPS methylation was inversely correlated with expression independently of ER status, and FDPS expression was highest in Basal and HER2-enriched intrinsic subtypes (Figure 10).

4.3 Subtypes, proliferation and immunity

The four subtypes recapitulate known breast cancer biology: C1 is predominantly ER-negative, proliferative (cell-cycle up) and immune-hot (highest CD8 T, regulatory T, NK, B-cell and macrophage scores); C3 is almost uniformly ER+, immune-cold and oxidative-phosphorylation-high; C2 shows down-regulated cell-cycle programs and high stromal/dendritic scores; C4 is a mixed, neutrophil-high group. Cholesterol-synthesis-high subtypes are thus luminal-like and immune-cold, whereas cholesterol-synthesis-low subtypes are more heterogeneous, proliferative and immune-active [2,8]. ssGSEA provided a rank-based, non-parametric complement confirming these patterns.

4.4 Strengths

All TCGA files were MD5-verified against the GDC manifest; external validation used independent cohorts and platforms; feature selection used two independent algorithms; subtype stability was assessed by consensus clustering with PAC; multiple-testing corrections were applied throughout; and the null results of the strict template were reported rather than engineered away.

4.5 Limitations

First, the strict WGCNA-by-pathway intersection was empty, and the candidate pool relied on the documented pathway-first adaptation. Second, an ER classifier built from expression partly rediscovers ER-pathway biology; its value is descriptive rather than mechanistic. Third, TCGA had only 122 PFI events, limiting subtype survival power; the METABRIC association ($P = 4.2 \times 10^{-5}$ univariable; $P = 0.013$ after ER/stage adjustment) is the stronger prognostic evidence, and subtype survival differences were not significant in GSE7390 (global log-rank $P = 0.48$). Fourth, single-cell localization is descriptive and observational, and the HPA protein evidence is limited to descriptive localization with a non-significant protein-level prognosis. Fifth, SMR used blood (eQTLGen) cis-eQTLs; breast-tissue instruments were unavailable for FDPS, and tissue-specific causal effects cannot be excluded. The CYP51A1 and LSS signals were HEIDI-rejected and should not be interpreted as causal; the FDPS HEIDI P-value (0.026) was not adjusted across probes, and GTEx v8 whole blood ($n = 670$) could not independently instrument FDPS. Coloc.abf PP.H4 = 0.9985 was not robust under SuSiE-based multi-signal colocalisation (maximum PP.H4 = 0.44; the SuSiE credible-set estimates did not fully converge with the external 1000 Genomes LD reference), so the shared-variant interpretation should be regarded as suggestive; conditional analysis confirmed additional GWAS-only signals without residual eQTL effects. The causal claim is confined to germline-regulated blood expression rather than tumour expression.

4.6 Causal inference By SMR with eQTLGen and BCAC summary statistics, causal inference was completed for 81 cholesterol-gene probes; FDPS passed Bonferroni correction and HEIDI, and Bayesian coloc.abf supported a shared causal variant at the FDPS locus (PP.H4 = 0.9985), together indicating a protective effect of genetically determined FDPS expression on breast cancer risk [43-45]. The protective direction was replicated in ER+ ($P = 1.54 \times 10^{-5}$, HEIDI $P = 1.09 \times 10^{-2}$) and ER- ($P = 2.78 \times 10^{-1}$) OncoArray sensitivity analyses [37], with ER+ reaching Bonferroni significance and ER- underpowered. These results are consistent with Mendelian-randomization evidence that statins reduce ER+ breast cancer risk [46] and with the established adjuvant benefit of nitrogen-containing bisphosphonates, which inhibit FDPS [47]. Multi-signal colocalisation was less conclusive (maximum PP.H4 = 0.44).

All single-cell, SMR/HEIDI and colocalisation pipelines are archived and reusable; follow-up work could refine the FDPS causal inference with independent fine-mapping resources, larger ER- GWAS and tissue-matched eQTLs.

4.7 Comparison with published lipid-metabolism signatures

Six published lipid-metabolism breast cancer signatures (2021-2025) were identified by systematic PubMed/Europe PMC search and compared on cohort use, signature size, immune deconvolution, subtype analysis, single-cell evidence and causal inference [7,48-52] (Supplementary Table S11). All six derived prognostic risk scores from TCGA-BRCA with LASSO-type feature selection; two validated across multiple independent cohorts, none performed germline causal inference (SMR/MR or Bayesian colocalisation), and none combined CIBERSORT/ESTIMATE deconvolution with external subtype mapping. To our knowledge, this is the first cholesterol-metabolism-focused breast cancer study to combine cross-platform ER-classifier validation, externally mapped molecular subtypes, immune deconvolution, single-cell localization and germline causal inference in one evidence chain.

5. Conclusions

Cholesterol metabolism genes do not form a progression-associated co-expression module in TCGA-BRCA, but they carry a strong, externally reproducible ER-status signal, define four biologically distinct immune/proliferation subtypes with PAM50 correspondence, localize to tumour epithelial and myeloid/endothelial compartments in single-cell data, and include FDPS as a putatively causal protective gene for breast cancer risk in SMR analysis. These results quantify both the value and the limits of the layered bioinformatics template for this pathway; the FDPS causal signal is supported by SMR and HEIDI in whole blood but should be interpreted cautiously given the multi-signal colocalisation results.

Declarations

Data availability. All data are public: TCGA-BRCA (NCI GDC; accessed 4-6 August 2026) and TCGA-CDR [14]; METABRIC (cBioPortal) [15]; PAM50 calls (UCSC Xena); FDPS HM450 methylation (cBioPortal); GSE21653 (NCBI GEO); GSE7390 (ArrayExpress E-GEOD-7390); GSE176078 (CELLxGENE); GSE161529 (Mendeley Data mirror of the Pal et al. atlas [12]); eQTLGen [35]; GTEx v8 [38,39]; BCAC GWAS (GWAS Catalog GCST010098/GCST010100/GCST004988; Zhang et al. 2020 [36] and Michailidou et al. 2017 [37]); 1000 Genomes Phase 3 EUR. Curated intermediate results are provided in the GitHub repository (<https://github.com/y0307Yd/brca-cholesterol-signature>) and the Zenodo archive (<https://doi.org/10.5281/zenodo.21873168>); raw data must be obtained from the original sources under their respective data-use policies. Supplementary Tables S1-S15 and Supplementary Figures S1-S3 accompany this manuscript; the TRIPOD-AI checklist (Supplementary Table S12), a comparison with published lipid-metabolism signatures (S11) and a graphical abstract are also provided in the repository.

Code availability. All analysis scripts (bc_01..bc_07, validation, single-cell, SMR/HEIDI, colocalisation, immune-deconvolution and bonus pipelines) and curated results are publicly available at GitHub (<https://github.com/y0307Yd/brca-cholesterol-signature>) and archived in Zenodo (<https://doi.org/10.5281/zenodo.21873168>).

Ethics. Secondary analysis of publicly available de-identified data; no new ethics approval was required. All data were used in accordance with the respective data-use policies of TCGA/GDC, METABRIC/cBioPortal, NCBI GEO and BCAC.

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Competing interests. The authors declare no competing interests.

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Figure Legends

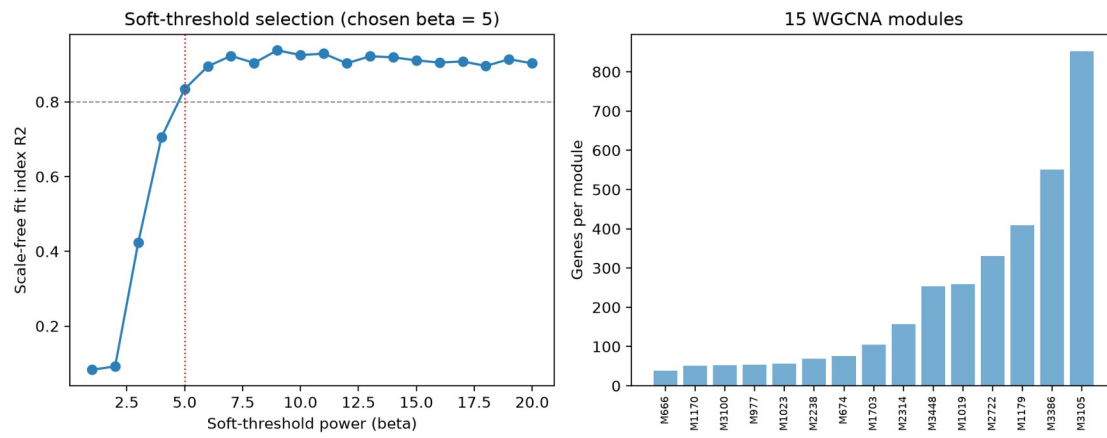


Figure 1. Soft-threshold selection and module sizes.



Figure 2. Module-trait correlation heatmap (asterisk: BH-FDR < 0.05).

ER-status classifier: calibration and decision-curve analysis

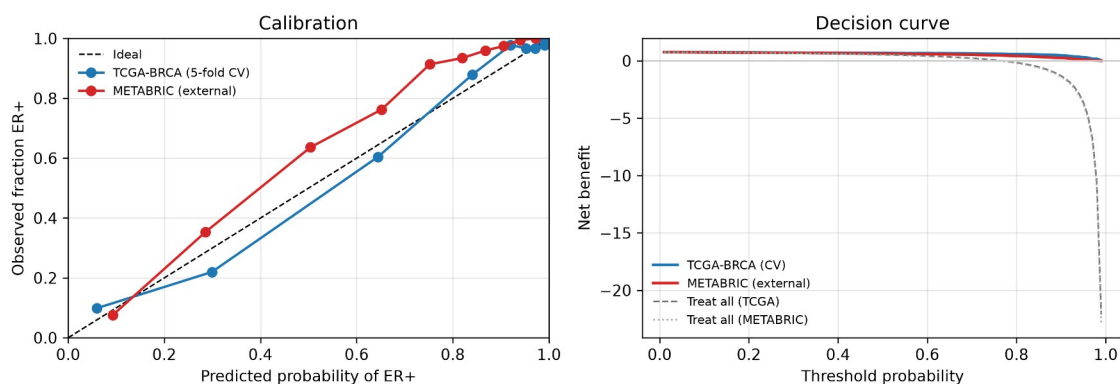


Figure 3. Calibration and decision-curve analysis of the 11-gene ER-status classifier. Left: reliability curves (deciles) for TCGA-BRCA out-of-fold cross-validated predictions and METABRIC external predictions. Right: decision curves showing net benefit versus threshold probability, with treat-all reference lines.

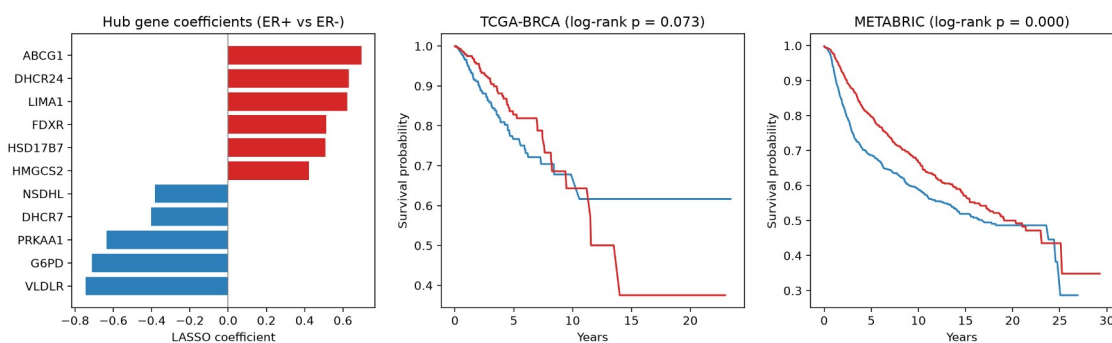


Figure 4. Hub-gene coefficients and Kaplan-Meier curves by signature median (TCGA-BRCA, METABRIC).

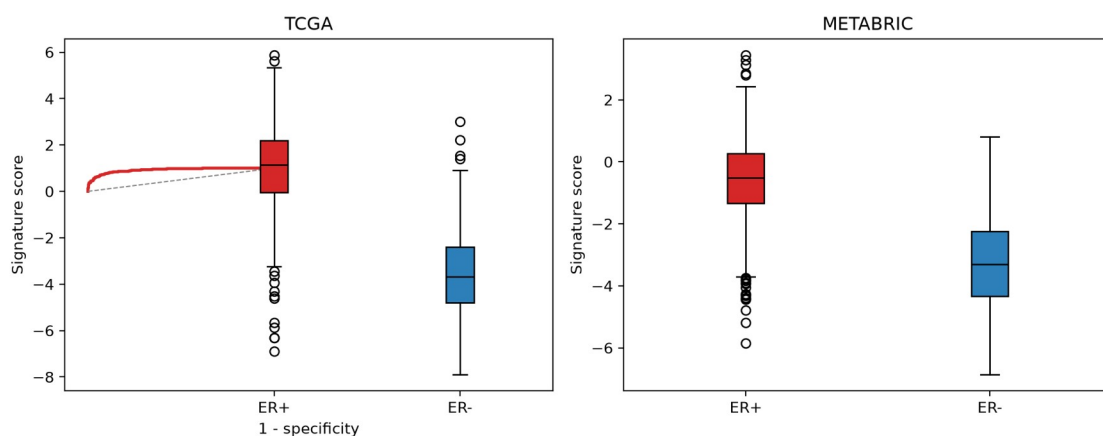


Figure 5. External ROC curve (METABRIC) and signature distribution by ER status.

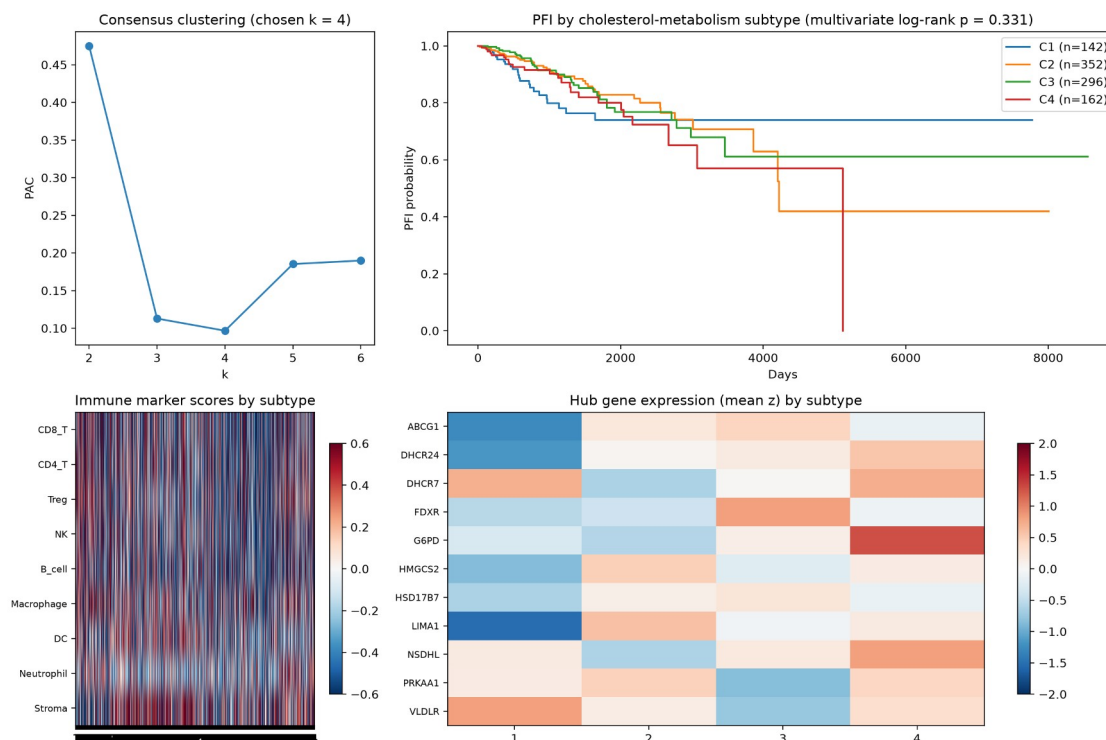


Figure 6. Consensus clustering PAC, subtype survival, immune scores and hub-gene heatmap.

Cross-cohort validation of cholesterol-metabolism subtypes

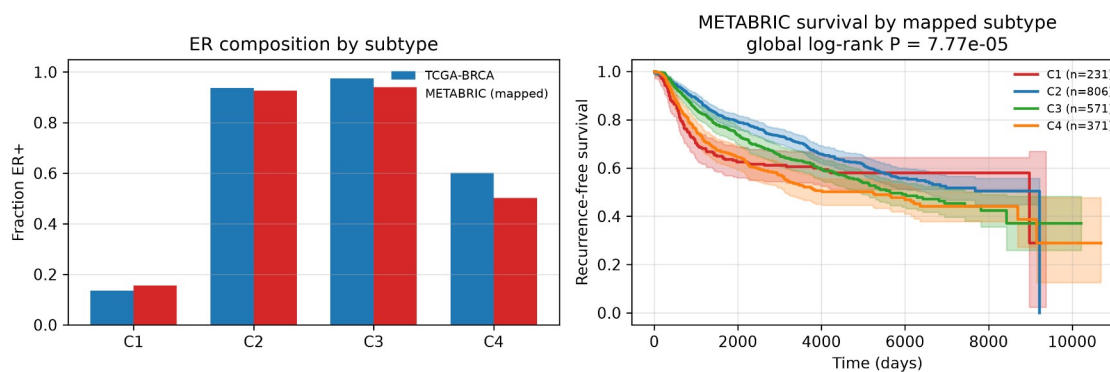


Figure 7. Cross-cohort validation of the four cholesterol-metabolism subtypes. Left: ER composition by subtype in TCGA-BRCA (consensus clustering) and METABRIC (nearest-centroid mapping). Right: Kaplan-Meier recurrence-free survival by mapped subtype in METABRIC (global log-rank P = 7.8e-05).

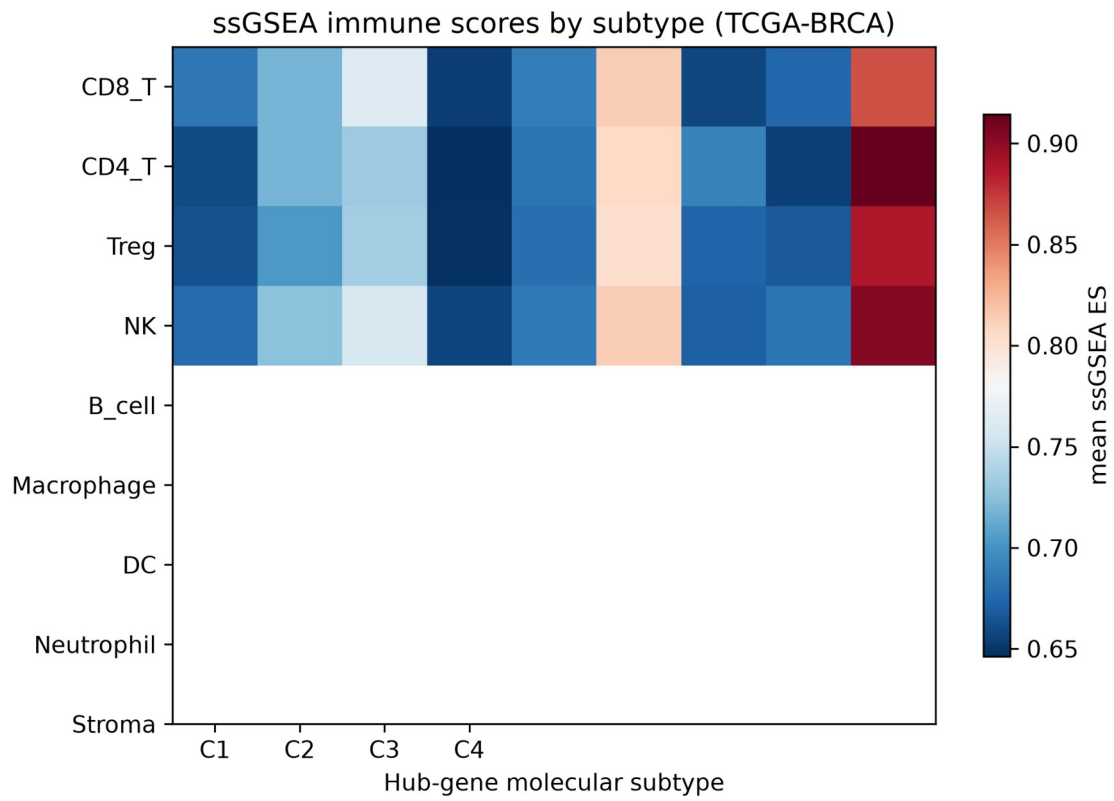


Figure 8. ssGSEA immune enrichment scores by hub-gene molecular subtype (TCGA-BRCA).

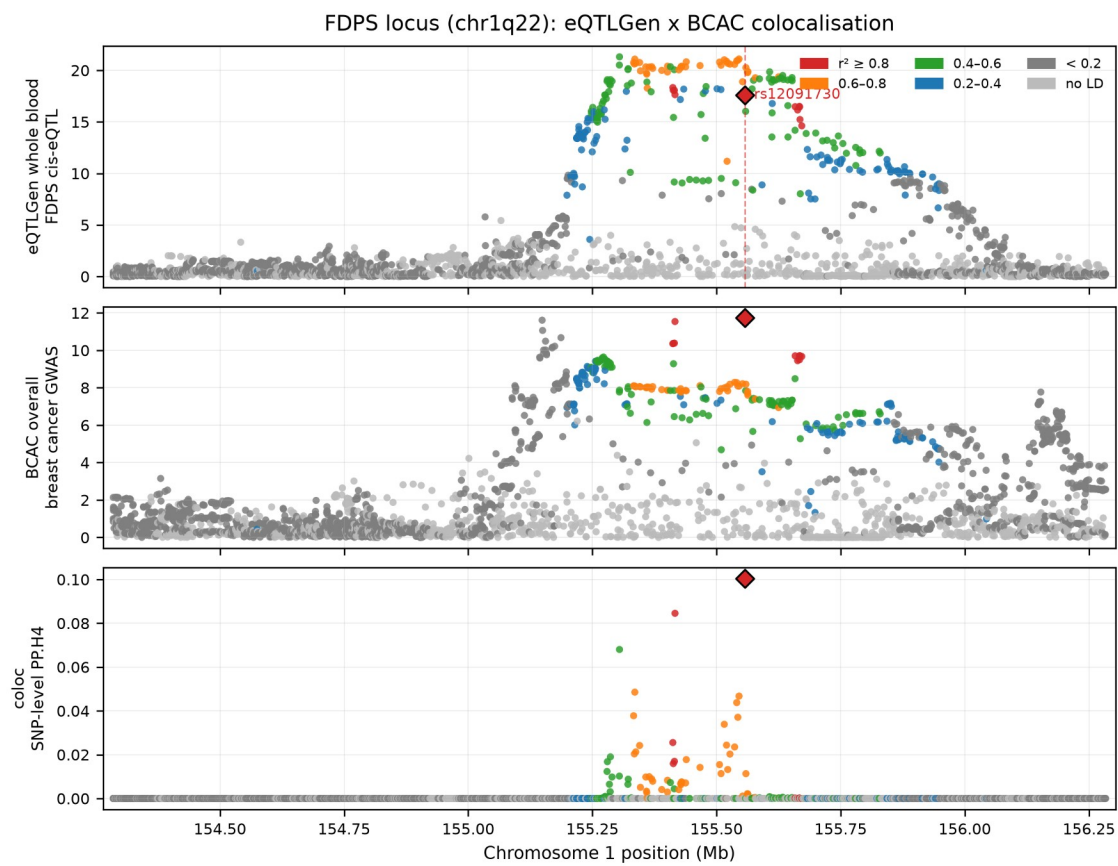


Figure 9. Colocalisation at the FDPS locus (eQTLGen whole-blood cis-eQTL versus BCAC overall breast cancer GWAS). Top: $-\log_{10} P$ for FDPS expression; middle: $-\log_{10} P$ for breast cancer risk; bottom: per-SNP posterior probability of a shared causal variant (PP.H4). Dots are coloured by LD (r^2) with the lead SNP rs12091730 in 1000 Genomes EUR. PP.H4 = 0.9985 over 3,248 shared SNPs.

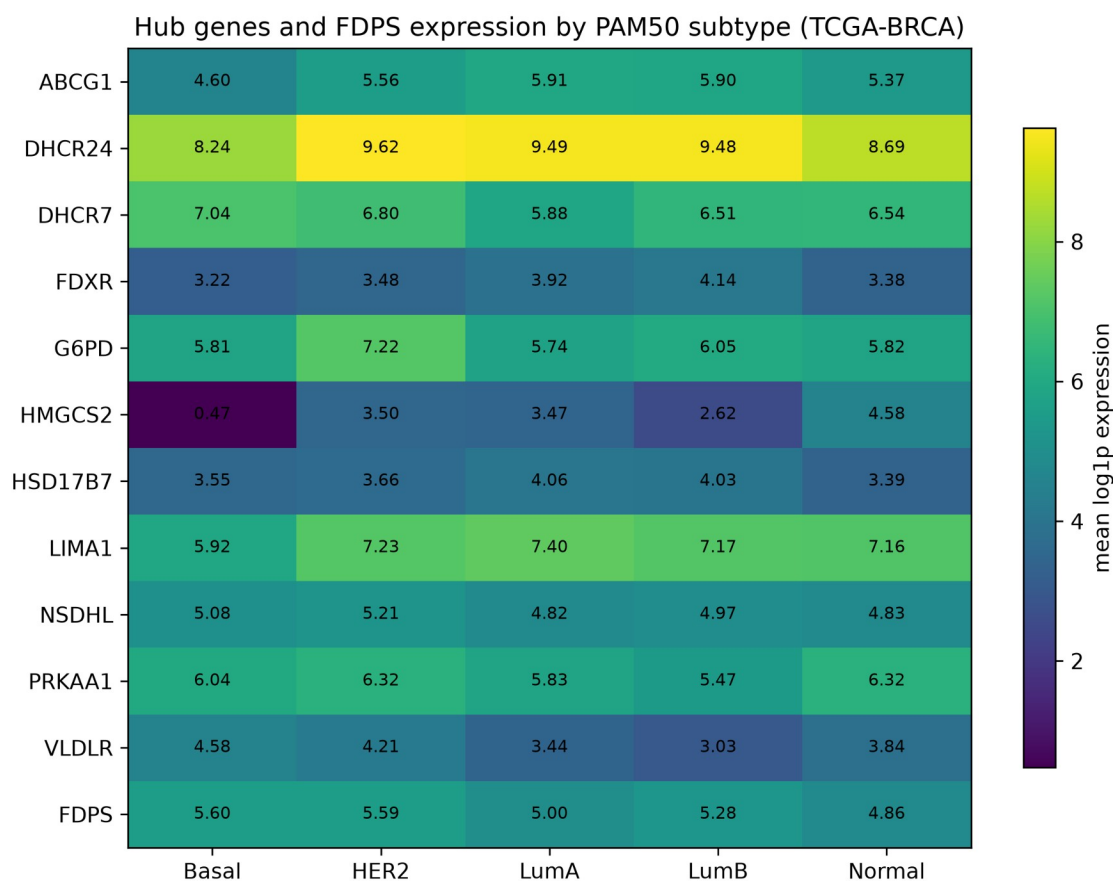


Figure 10. Hub genes and FDPS expression by PAM50 intrinsic subtype (TCGA-BRCA). Heat-map of mean log1p expression.

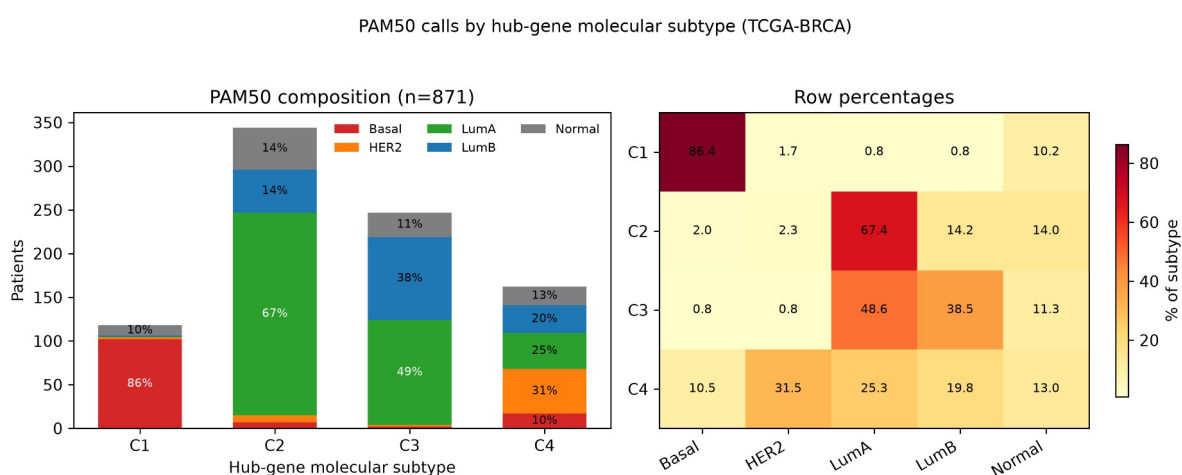


Figure 11. PAM50 composition of the four hub-gene molecular subtypes in TCGA-BRCA. Left: stacked bars showing the number of patients per PAM50 call within each hub-gene subtype; right: heat-map of row percentages.

GSE7390 independent validation (n=198)

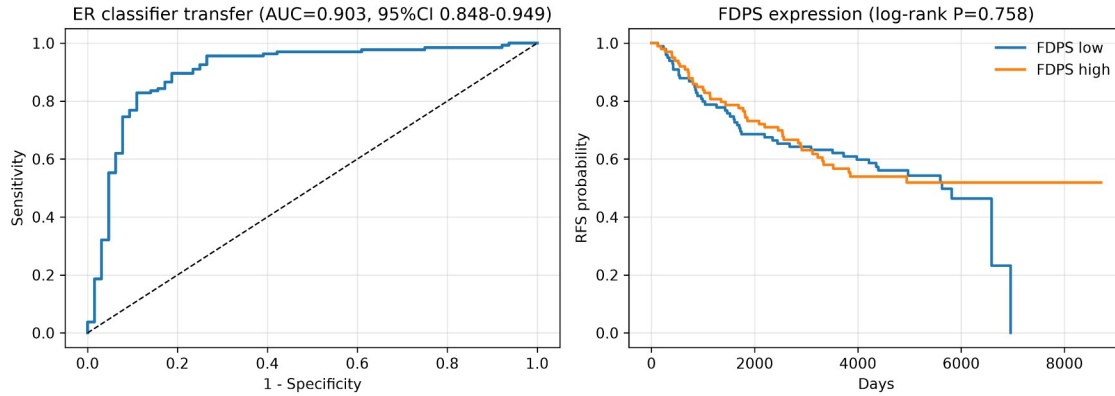


Figure 12. Independent validation of the ER classifier and FDPS in GSE7390 (Affymetrix U133A). Left: ROC curve of the transferred 11-gene ER classifier against IHC-defined ER status; right: recurrence-free survival by median FDPS expression.

Independent validation: GSE21653 (n=252, events=83)

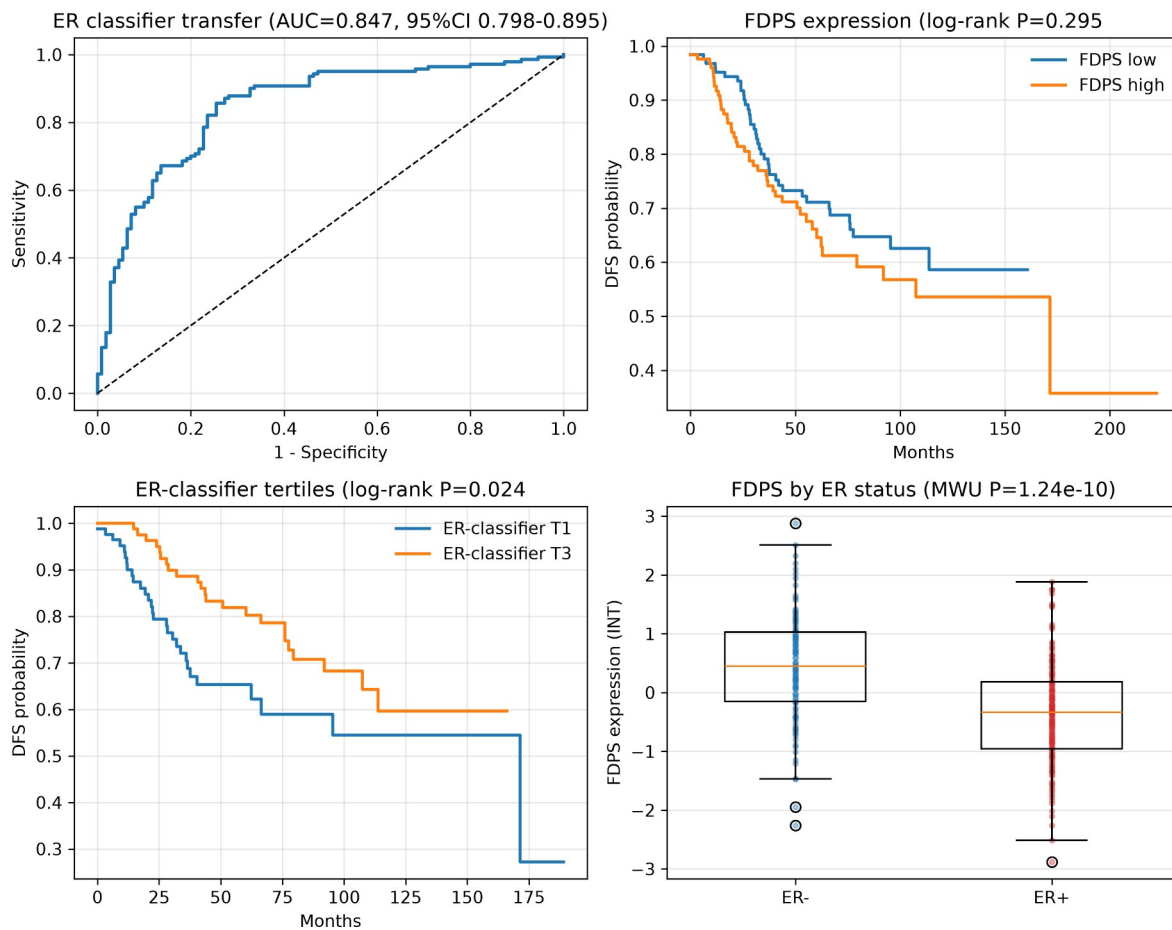


Figure 13. Independent validation of the ER classifier and FDPS in GSE21653. Top left: ROC curve of the transferred 11-gene ER classifier against immunohistochemistry-defined ER status; top right: disease-free survival by median FDPS expression; bottom left: disease-free survival by ER-classifier tertiles; bottom right: FDPS expression by ER status.

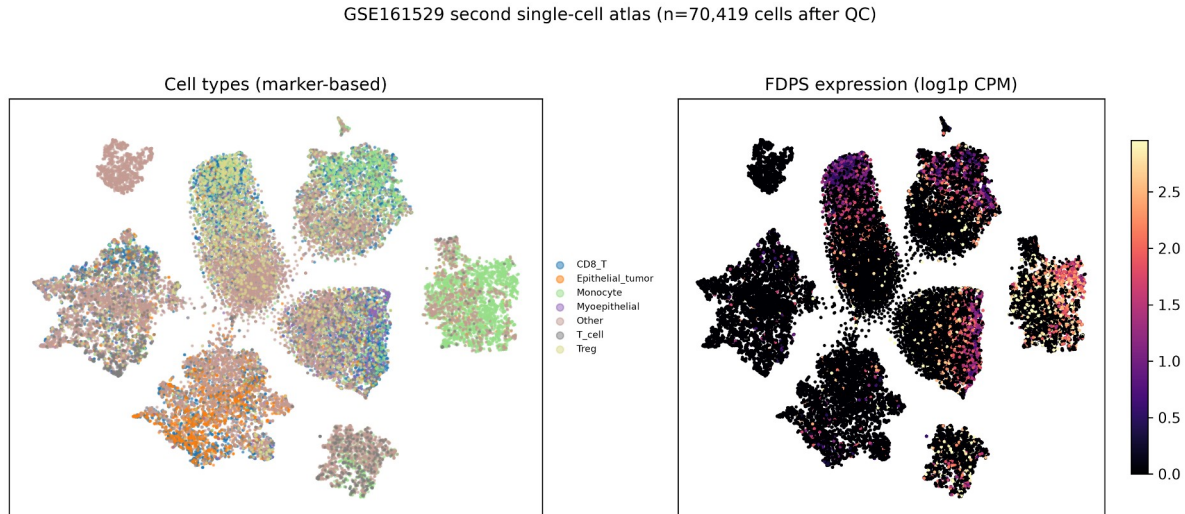


Figure 14. Second single-cell atlas (GSE161529). Left: t-SNE embedding coloured by marker-based cell types; right: FDPS expression overlaid on the same embedding.

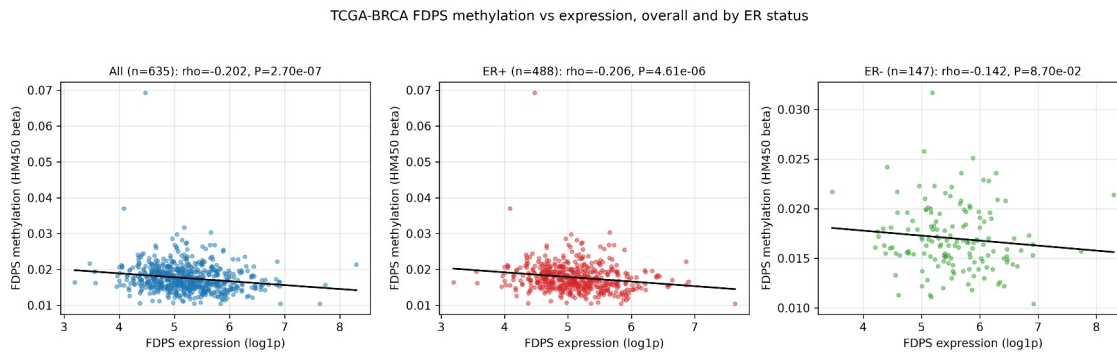


Figure 15. FDPS DNA methylation versus expression in TCGA-BRCA, overall and by ER status (cBioPortal HM450).

TCGA-BRCA: FDPS DNA methylation vs expression (cBioPortal HM450)

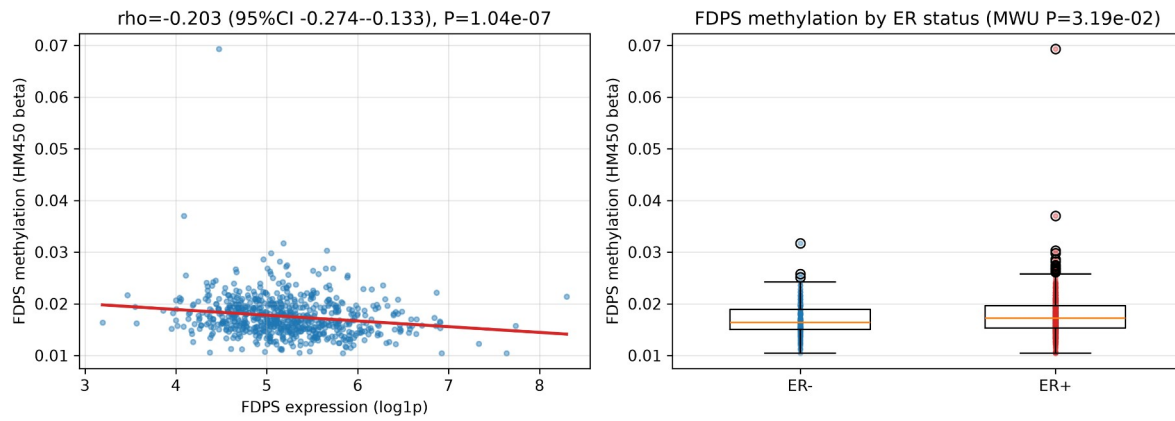


Figure 16. FDPS DNA methylation versus FDPS expression in TCGA-BRCA (cBioPortal HM450). Left: scatter plot of mean FDPS-associated methylation beta against FDPS expression; right: methylation beta by ER status.